MUSKAN KARNANI

Senior Software Engineer

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Professional Summary

Backend developer with 5+ years of experience building enterprise-scale systems at HSBC, specializing in Java, Spring Boot, and microservices architecture. Led the transformation of a legacy IBM reporting platform into microservices, boosting performance by 40%, and modernized core services from Java 8 to Java 21, cutting technical debt by 30%. Skilled in Agile delivery, production support, and system design, with a consistent track record of replacing costly vendor tools with in-house solutions that reduce operational costs and improve platform flexibility.

Skills

Java • Java 21 • Spring Boot • Spring Framework • REST APIs • Microservices • Oracle SQL • AWS S3 • Jenkins • GitHub • Linux • Shell scripting • CI/CD • Agile • Production Support • System Design • Continuous integration & deployment • Microservices architecture • Design reviews • Project leadership

Experience

HSBC Pune, India

Senior Software Engineer *Nov 2023 – Present*

- Directed transformation of legacy reporting platform into microservices architecture, boosting scalability and **system performance by 40%** while allowing decommissioning of vendor-based IBM reporting platform
- Architected and executed upgrade from Java 8 to Java 21 along with modernization of Spring 2.x services, **reducing technical debt by 30%** and enabling faster and more secure feature deliver
- Redesigned and migrated a Python-based data processing utility to Java 21, **reducing processing time by 50%** and improving reliability, maintainability, and integration with enterprise systems
- Designed and developed an in-house Swagger/OpenAPI-based reporting and business rule validation solution to replace the legacy vendor tool, **reducing operational costs by 40%** and **eliminating $10.60–$11.25** in per-user monthly licensing fees
- Spearheaded architecture analysis and engineered an AWS S3-based storage solution, **eliminating Oracle database licensing costs** (up to **$47,500** per processor or **$950** per Named User Plus) and enabling real-time business calculations directly from input files
- Led Agile delivery activities including sprint planning, technical design discussions, and code reviews, **improving release cycle speed by 25%**
- Produced 12 technical design documents, accelerating team access to critical information and **reducing onboarding time by 25%** and **reduced support requests by 30%**
- Managed production support and BAU activities, **achieving 99.9% system uptime** and addressing critical performance bottlenecks

HSBC Pune, India

Software Engineer *Sep 2020 – Nov 2023*

- Delivered an in-house Wolf Management System, replacing Fircosoft Business Utilities; this initiative improved operational control and **reduced vendor dependency by 100%**
- Enhanced WMS functionalities including audit logging, deployment scheduling, and server refresh modules, elevating operational efficiency and **enhancing user experience by 40%**
- Rebuilt the cross-regional data extractor, **achieving a 75% reduction in data retrieval errors** and ensuring high-integrity data for critical business decisions.
- Managed CI/CD deployments using Jenkins and Ansible, supporting at least 5 releases per sprint with seamless integration, automation, and reliable delivery
- Provided production support and system monitoring using E2 and Control-M as part of an on-call rotation, resolving critical incidents and maintaining operational stability with minimal recurring issues

IIT Delhi Delhi, India

Project Intern *2019*

- Engineered underwater object detection system utilizing depth images obtained from Microsoft Kinect V2
- Implemented background subtraction and k-means clustering algorithms to eliminate noise and forecast object trajectories from depth images, , improving detection accuracy by 15%

Accomplishments

- **Processathon Winner (2025) - Built an ICE score validation feature eliminating dependency on legacy internal APIs**
- **Google Cloud Associate Cloud Engineer (2024)**
- **HerHack Winner (with Google, 2023)** – Innovated an AI-powered MoM platform using cloud services, Python, and Angular
- **Enterprise Engineer Program (EEP), HSBC** (2023)
- **HackForGood Participant (2023)** – Created a language translator processing utility

Education

Banasthali Vidyapith
Bachelor of Technology in Computer Science – CGPA: 8.43 *2020*
Lucknow Public School
Senior and Higher Secondary (ISC) – 96% *2016*